



1.2 Two-Track Structure

This scholarship is offered as two distinct tracks, described in detail in Chapter 2:

1. **Need-Based Scholarship** — for applicants who can demonstrate financial hardship that is a barrier to MCAT preparation.
2. **Merit-Based Scholarship** — for applicants who may not qualify under the need-based criteria but demonstrate strong academic dedication and commitment to medicine.

1.3 Scholarship Summary

Award Name	The M-CATs Registration Scholarship
Award Amount	M-CAT Registration Cost
Number of Awards	Three (3) needs based & two (2) merit based
Award Cycle	Annual



2.2 Merit-Based Scholarship

This track applies to applicants who may not qualify under the need-based criteria but demonstrate strong academic dedication and commitment to medicine.

2.2.1 Application Essay Questions

1. (250–500 words) The journey to becoming a med student is full of obstacles and unplanned circumstances. Upon studying for the MCAT you will face such challenges. Using your previous experience, how do you plan on approaching and overcoming any future unexpected problems during your future MCAT studies?
2. (250–500 words) How have you demonstrated academic dedication and commitment to medicine (e.g., engagement in M-CATS seminars, peer tutoring, or coursework)?

2.2.2 Merit Verification Methods

- **Unofficial Transcript:** Accept an unofficial transcript screenshot or PDF directly from the student portal (myUK) showing cumulative GPA and pre-med science coursework, avoiding official transcript fees.

2.3 Need-Based Scholarship

This is designed for applicants who can demonstrate genuine financial hardship.

2.3.1 Application Essay Questions

1. All of the essays required for the merit based track are required here. Apart from 2.2.2 (an Unofficial Transcript)
2. **Statement of Financial Hardship** (250–500 words): A short written statement detailing uncompensated financial burdens or changes in circumstances.
 - *(Optional)* Provide any evidence of financial hardship (e.g., Pell Grant eligibility, pay stub, FAFSA record, tax return). Documents included here can be referenced in the Statement of Financial Hardship and will further educate the committee on the applicant's financial background.

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3. **Impact Statement** (250–500 words): Describe how receiving this award will relieve these barriers.



3.1 Ongoing Requirements

- **Required progress checks / one-on-one meetings** with an officer of the board to ensure success throughout the MCAT preparation timeline.

3.2 Conditional Acceptance and Disbursement

Awardees receive **conditional acceptance** into the scholarship program. Under this structure:

- Students who receive this scholarship will be reimbursed or provided funding for MCAT registration **once MCAT board members have determined** that they display the competence and commitment necessary to achieve success on the MCAT, with respect to their planned timeline and target score.

Gen Chem 4

compounds and stoichiometry

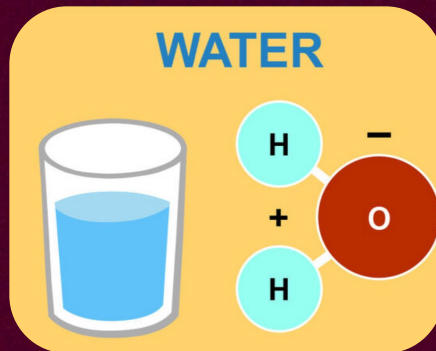
M-CATS @ university of kentucky
9/25 chem session 6:00 PM

-I-

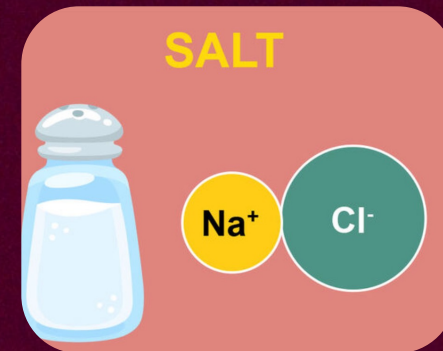
If are we counting

Compounds

Molecular Compounds

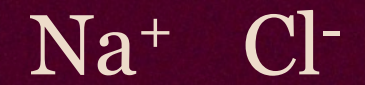
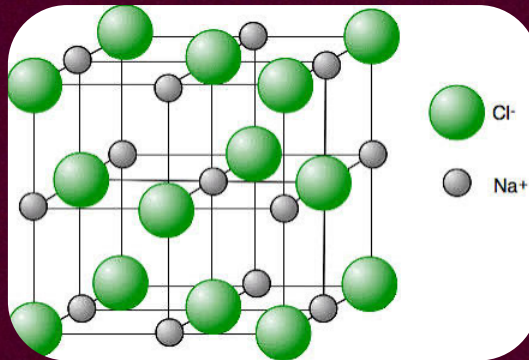
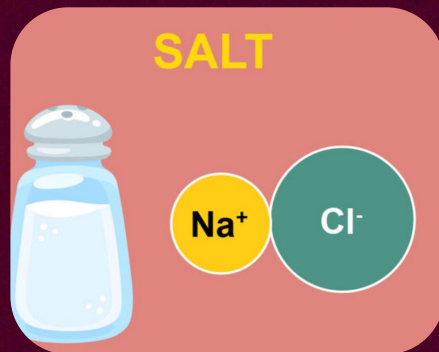


Ionic Compounds



Compounds

Ionic Compounds



Weight

Molecular Weight

Formula Weight

Moles

-II-

How do we describe a compound

Formulas

Molecular

Empirical

Structural

% composition

H₂O?

Empirical Formula from %

Imagine a compound is 40.0% C, 6.7% H, and 53.3% O.

Procedure

1. Assume 100g of compound
2. Convert to mols
3. Divide by smallest mole amount

The Molar mass of the molecular compound is 60 g/mol

-III-

Types of chemical rxns
also balancing btw

Subscripts versus coefficients

Synthesis

Double Replacement

Decomposition

Combustion

Single Replacement

Neutralization

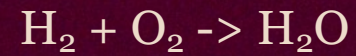
Balancing

-IV-

How much reacts

Stoichiometry

Grams $\rightarrow$ moles $\rightarrow$ mole ratio $\rightarrow$ moles $\rightarrow$ grams

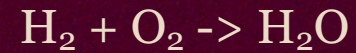


I have 8.0 g of H_2 , how many grams of water will I have?

Limiting Reagent

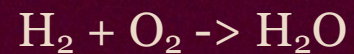
2 bread + 1 cheese $\rightarrow$ 1 sandwich

I have 1e8 bread and 2 cheese. How many sandwiches?



I have 8.0 g of H_2 and 16.0 g of O_2 , how many grams of water will I have?

Yield



I have 8.0 g of H_2 and 16.0 g of O_2 , how many grams of water will I have?

Theoretical Yield

Experimental/Actual Yield

% Yield

-IV-

On Fans

Gram-Equivalent Weight

Polyatomic Ions

Equivalents

Common Charges

Normality

Transition Metals

Electrolytes

-V-

Questions

A sample contains 36.0g of H₂O

- a) What is the molar mass?
- b) How many moles of H₂O are present?
- c) Approximately how many H₂O molecules are present?

A compound is found to contain:

- 52.2% carbon
- 13.0% hydrogen
- 34.8% oxygen

Its molar mass is approximately 138 g/mol.

- a) Determine empirical formula
- b) Determine molecular formula

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- a) Determine empirical formula
- b) Determine molecular formula

Also don't worry
About whether this compound
Is chemically neutral :)
I lowkey just made it up, so don't
@ me

A student burns 15.0g of propane, C_3H_8 with 50.0g of oxygen gas.

Given molar masses:

$M(\text{C}_3\text{H}_8) = 44.0 \text{ g/mol}$

$M(\text{O}_2) = 32.0 \text{ g/mol}$

$M(\text{CO}_2) = 44.0 \text{ g/mol}$

$M(\text{H}_2\text{O}) = 18.0 \text{ g/mol}$

- a) Balance the equation
- b) Determine the limiting reactant
- c) Calculate the theoretical yield
- d) If the student actually obtains 35.0g of CO_2 , calculate the percent yield